

## Project Initialization and Planning Phase

|                      |                                                        |
|----------------------|--------------------------------------------------------|
| <b>Date</b>          | June 28, 2026                                          |
| <b>Team ID</b>       |                                                        |
| <b>Project Title</b> | Credit Card Approval Prediction using Machine Learning |
| <b>Maximum Marks</b> | 5 Marks                                                |

### Project Proposal (Proposed Solution) Report

This project automates the credit card approval process using machine learning. It analyses applicant financial and demographic information to predict whether a credit card application should be approved or rejected.

| <b>Project Overview</b>  |                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective                | Develop an intelligent machine learning system that predicts whether a credit card application should be approved or rejected based on applicant information with high accuracy.                                                                                                                                                                                                    |
| Scope                    | The project includes data collection, exploratory data analysis (EDA), data preprocessing, feature engineering, model training, model evaluation, comparison of multiple machine learning algorithms, and deployment of the best-performing model using a Flask web application for real-time credit card approval prediction.                                                      |
| <b>Problem Statement</b> |                                                                                                                                                                                                                                                                                                                                                                                     |
| Description              | Banks receive thousands of credit card applications every day. Manual verification is time-consuming, inconsistent, and prone to human error, resulting in delayed approvals and increased operational costs.                                                                                                                                                                       |
| Impact                   | Automating the approval process improves decision-making speed, increases prediction accuracy, reduces financial risks, minimizes manual effort, and enhances customer satisfaction.                                                                                                                                                                                                |
| <b>Proposed Solution</b> |                                                                                                                                                                                                                                                                                                                                                                                     |
| Approach                 | Historical applicant data is preprocessed and used to train multiple machine learning models including Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. The best-performing model is integrated into a Flask web application for real-time credit card approval prediction.                                                                                |
| Key Features             | <ul style="list-style-type: none"> <li>• Automated Credit Card Approval Prediction</li> <li>• Data Preprocessing and Feature Engineering</li> <li>• Multiple Machine Learning Model Comparison</li> <li>• Real-Time Prediction using Flask</li> <li>• User-Friendly Web Interface</li> <li>• Fast and Consistent Decision-Making</li> <li>• Responsive HTML/CSS Frontend</li> </ul> |

## Resource Requirements

| Resource Type           | Description                                                    | Specification/ Allocation                                                                                                                                                                                                           |
|-------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hardware</b>         |                                                                |                                                                                                                                                                                                                                     |
| Computing Resources     | CPU/GPU specifications, number of cores                        | Intel Core i5/i7 Processor (4+ Cores)                                                                                                                                                                                               |
| Memory                  | RAM specifications                                             | 8 GB RAM (Minimum), 16 GB Recommended                                                                                                                                                                                               |
| Storage                 | Disk space for datasets, trained models, and application files | 512 GB SSD / 1 TB HDD                                                                                                                                                                                                               |
| <b>Software</b>         |                                                                |                                                                                                                                                                                                                                     |
| Frameworks              | Python Web Framework                                           | Flask                                                                                                                                                                                                                               |
| Libraries               | Machine Learning and Data Processing Libraries                 | Scikit-learn<br>Pandas<br>NumPy<br>Matplotlib<br>Seaborn<br>Flask<br>Joblib                                                                                                                                                         |
| Development Environment | IDE                                                            | Google Colab<br>Visual Studio Code                                                                                                                                                                                                  |
| <b>Data</b>             |                                                                |                                                                                                                                                                                                                                     |
| Data                    | Source, Size, Format                                           | Source:<br>Kaggle Credit Card Approval Dataset<br><br>Files:<br>application_record.csv<br>credit_record.csv<br><br>Format:<br>CSV<br><br>Final Dataset:<br>Approximately 219,173 processed records after merging and preprocessing. |